

6: Bayesian decision analysis and practical significance with ROPE, emmeans, and marginaeffects

Bayesian Mixed Effects Models with brms for Linguists

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1 The Problem: Does Effect X Matter?

1.1 Research Scenario

Imagine you’ve run a psycholinguistic experiment and fitted a Bayesian model. You have posterior distributions for your effects. Now you face the question:

“Is this effect meaningful in practice?”

- You have an estimate (e.g., $\delta = 0.12$ log-RT units)
- You have uncertainty (95% CI: [0.08, 0.16])
- But: Is this difference big enough to matter?

Practical significance is different from **statistical significance**:

- Statistical: “Is there an effect?” (distinguishable from noise) (we will discuss this in the session on Bayes Factor)
- Practical: “Does it matter?” (large enough to care about)

1.2 Why Practical Significance Matters

Consider these scenarios:

Scenario A: Statistically credible but trivial

- Effect: 2ms difference in reading time
- 95% CI: [1ms, 3ms] (clearly non-zero!)
- But: Readers can’t perceive < 20 ms differences
- **Conclusion:** Real effect, but irrelevant

Scenario B: Uncertain but possibly large

- Effect: 50ms difference (noticeable to readers)
- 95% CI: [-10ms, 110ms] (wide uncertainty)
- Traditional conclusion: “not significant”
- But: Most credible values are meaningful
- **Conclusion:** Worth investigating further

2 Where We Are in the Analysis Workflow

2.1 The Bayesian Workflow So Far

Let’s review where we’ve been:

1. **Module 01:** Set priors
2. **Module 02:** Prior predictive checks
3. **Module 03:** Posterior predictive checks
4. **Module 04:** Compare priors $\rightarrow$ Sensitivity analysis
5. **Module 05:** Compare models $\rightarrow$ Evaluate predictive performance (LOO)
6. **Today (Module 06):** Practical significance (ROPE + comparisons)
7. **Module 07:** Bayes factors & hypothesis comparison
8. **Module 08:** Convergence

2.2 When to Use Each Approach

QUESTION	TOOL
Which model structure is better?	LOO (Module 05)
Is effect meaningful?	ROPE (Module 06)
Compare multiple groups?	emmeans (Module 06)
Custom predictions/contrasts?	marginaleffects (M06)
Evidence for hypothesis?	Bayes Factor (M07)
Are estimates robust to priors?	Prior comparison (M04)

2.3 Choosing The Right Tool

Use **ROPE** when:

- You want to declare “effect too small to matter”
- You need clear decision rules (accept/reject/undecided)

Use **emmeans** when:

- You have factorial designs (multiple groups/conditions)

Use **marginaleffects** when:

- You’re working with complex models (GAMs, interactions)

Use **Bayes Factors (Module 07)** when:

- You want to quantify evidence for one hypothesis over another (see Module 07)

Use **LOO (Module 05)** when:

- Comparing different model structures
- Doing feature selection
- Predictive performance is primary concern

2.4 Setup

2.5 Generate Reaction Time Data

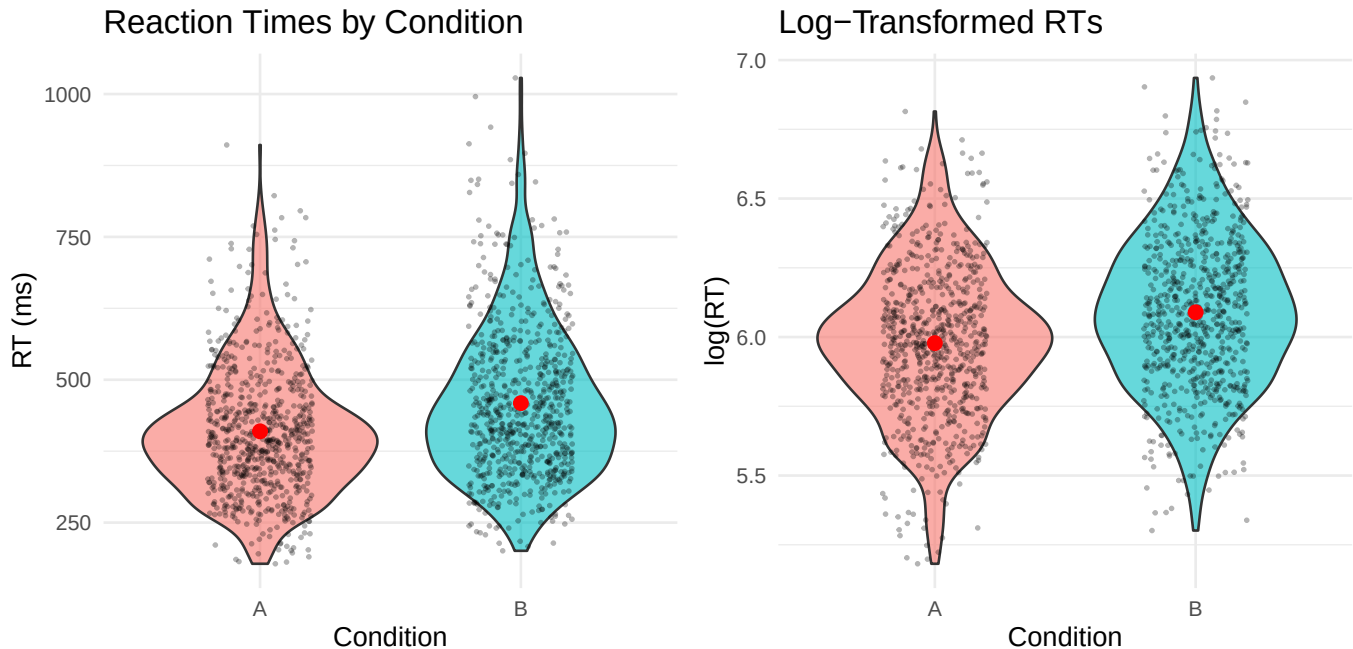
We’ll generate RT data similar to previous modules, but with specific properties useful for hypothesis testing demonstrations:

- Clear directional effect (Condition B slower than A)
- Effect size in a realistic range for psycholinguistics
- Adequate sample size for stable estimates

```
# A tibble: 2 x 6
  condition     n mean_rt median_rt sd_rt mean_log_rt
  <fct>      <int>   <dbl>    <dbl> <dbl>      <dbl>
1 A           720     410      395    114        5.98
2 B           720     459      439    133        6.09
```

```
# A tibble: 1 x 2
  Measure      Value
  <chr>      <dbl>
1 Effect size (log scale) 0.111
```

2.6 Visualize the Data



2.7 Fit the Model

2.8 Model Summary

```
Family: gaussian
Links: mu = identity
Formula: log_rt ~ condition + (1 + condition | subject) + (1 | item)
Data: rt_data (Number of observations: 1440)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

```
~item (Number of levels: 24)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.11    0.02    0.08    0.16 1.00     870    1458
```

```
~subject (Number of levels: 30)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
sd(Intercept)    0.17    0.02    0.13    0.22 1.00     790
sd(conditionB)    0.06    0.02    0.02    0.09 1.00     879
cor(Intercept,conditionB) 0.03    0.26   -0.45    0.54 1.00    2095
Tail_ESS
```

```
sd(Intercept)          1296
sd(conditionB)          726
cor(Intercept,conditionB) 2035
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	5.98	0.04	5.90	6.05	1.01	478	992
conditionB	0.11	0.02	0.08	0.14	1.00	2917	2927

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.20	0.00	0.19	0.21	1.00	4011	3063

Draws were sampled using `sample(hmc)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat = 1`).

3 ROPE (Kruschke, 2015) and Decision Analysis (Gelman et al., 2013)

3.1 ROPE

Statistical significance tells us if an effect differs from zero. But:

- **Statistical question:** Is the effect *exactly* zero?
- **Practical question:** Is the effect *close enough* to zero to ignore?

This is where **ROPE** (Region of Practical Equivalence) comes in.

3.1.1 Making ROPE-based Decisions

Instead of testing if $\theta = 0$ exactly, we define a small interval around zero:

$$\text{ROPE} = [-\varepsilon, +\varepsilon]$$

where ε is the **smallest effect size we care about** (domain-specific).

Decision rules:

1. **95% HDI entirely inside ROPE** $\rightarrow$ Accept practical equivalence (effect negligible)
2. **95% HDI entirely outside ROPE** $\rightarrow$ Reject equivalence (effect matters)
3. **95% HDI overlaps ROPE** $\rightarrow$ Uncertain (collect more data or accept uncertainty)

3.1.2 Setting ROPE Boundaries

! Common Pitfall: Post-Hoc ROPE Boundaries

ROPE boundaries must be set **before seeing results** based on:

- Domain knowledge (e.g., “RT differences < 50ms are imperceptible”)
- Standardized effect sizes (e.g., “Cohen’s $d < 0.1$ considered negligible”)

Mistake: Setting ROPE after looking at posterior to get desired conclusion.

Why it matters: Post-hoc boundaries invalidate the test (like p-hacking).

3.1.3 Four Methods for Justifying ROPE Boundaries

From Kruschke (2015, Chapter 12): ROPE boundaries should represent the smallest effect you care about. Here are four principled methods for setting them:

3.1.3.1 Method 1: Previous Research / Meta-Analysis

Use effect sizes from prior literature to calibrate what counts as “small.”

Method	Typical Effect	ROPE Boundaries	Interpretation
Based on Meta-Analysis	0.10 log-units (100ms or 10%)	$[-0.03, 0.03]$	Effects < 3% are small relative to field norms

When to use:

- Mature research area with existing effect size estimates
- You want to compare to “typical” effects in the field
- You have access to meta-analyses or large-scale studies

Example:

In their meta-analysis of 50 reading time studies, Smith et al. (2020) report a mean effect of $d = 0.35$ for syntactic complexity manipulations. We set our ROPE at $d = 0.10$ (approximately 1/3 of the typical effect), corresponding to ± 0.03 log-RT units.

3.1.3.2 Method 2: Measurement Precision

ROPE should exceed measurement error—otherwise you’re testing noise.

Method	Measurement Error	ROPE Boundaries	Interpretation
Based on Measurement Precision	0.02 log-units (20ms)	$[-0.03, 0.03]$	Effects smaller than measurement error are unreliable

When to use:

- You have reliability/measurement error estimates
- You want to avoid claiming effects smaller than noise
- Measurement precision is the limiting factor

How to estimate measurement error:

```
# A tibble: 3 x 2
  Measure      Value
  <chr>      <chr>
1 Within-subject SD 0.085
2 Suggested ROPE (±) 0.127
3 ROPE Boundaries  [-0.127, 0.127]
```

3.1.3.3 Method 3: Perceptual/Practical Thresholds

Effects below perception or practical impact are negligible by definition.

Method	Perceptual Threshold	ROPE Boundaries	Interpretation
Based on Perceptual Threshold	50ms (0.05 log-units)	[-0.05, 0.05]	Effects imperceptible to readers are negligible

When to use:

- You can measure perceptual/practical thresholds
- You care about real-world impact (not just statistical detection)
- You have pilot data or existing benchmarks

Example applications:

- **Reading:** Just-noticeable difference in reading time
- **Accuracy:** Minimal detectable accuracy change
- **Rating scales:** Smallest perceived difference on Likert scale
- **Medical:** Minimum clinically meaningful difference (MCID)

How to measure perceptual thresholds:

Pilot study design:

1. Show participants trials from both conditions
 2. Ask: "Did you notice any difference in difficulty/speed?"
 3. Correlate subjective reports with actual RT differences
 4. Find threshold below which participants don't notice
- Use this threshold as ROPE

3.1.3.4 Method 4: Standardized Effect Sizes (Cohen's d)

Use conventional effect size benchmarks from your field.

Approach: Cohen's guidelines suggest $d = 0.2$ is a "small" effect. We'll set ROPE at half of that: $d = 0.10$, representing a "very small" effect.

To convert Cohen's d to the log-RT scale, we use:

$$d = \frac{\mu_1 - \mu_2}{SD_{\text{pooled}}}$$

Formula for pooled standard deviation:

$$SD_{\text{pooled}} = \sqrt{\frac{\sum_{i=1}^k (n_i - 1) \times SD_i^2}{\sum_{i=1}^k (n_i - 1)}}$$

where k is the number of groups (conditions), n_i is the sample size for group i , and SD_i is the standard deviation for group i .

Cohen's d Threshold	Pooled SD (log-units)	ROPE Lower	ROPE Upper	Interpretation
0.1	0.278	-0.028	0.028	Effects with $d < 0.10$ are 'very small'

When to use:

- No domain-specific benchmarks available
- You want to follow field conventions
- Standardized metrics are expected in your area

Common benchmarks:

- **Cohen's d:** Small = 0.2, Medium = 0.5, Large = 0.8
- **Correlation (r):** Small = 0.1, Medium = 0.3, Large = 0.5
- **R²:** Small = 0.01, Medium = 0.09, Large = 0.25

Note: These are **conventions**, not laws of nature! Domain-specific thresholds (Methods 1-3) are usually better.

3.1.4 For RT Data (log-scale)

For our analysis:

- **ROPE** = [-0.05, +0.05] on the log scale
- **On original scale:** This corresponds to RT ratios between 0.95 and 1.05
 - Since we model $\log(\text{RT})$, a difference of ± 0.05 on the log scale = $e^{\pm 0.05}$ = multiplying RT by 0.95 to 1.05
 - Example: If Condition A = 400ms, ROPE means effects between 380ms and 420ms ($\pm 5\%$)
- **Interpretation:** RT differences smaller than 5% are too small to be practically meaningful

3.2 Understanding ROPE Graphically

3.2.1 Excursion to Decision Analysis: The Implicit Utility Function (Gelman et al., 2013)

Let's make explicit what ROPE boundaries mean in decision-theoretic terms (Gelman et al., 2013, Chapter 9; [Stan User's Guide on Decision Analysis](#)):

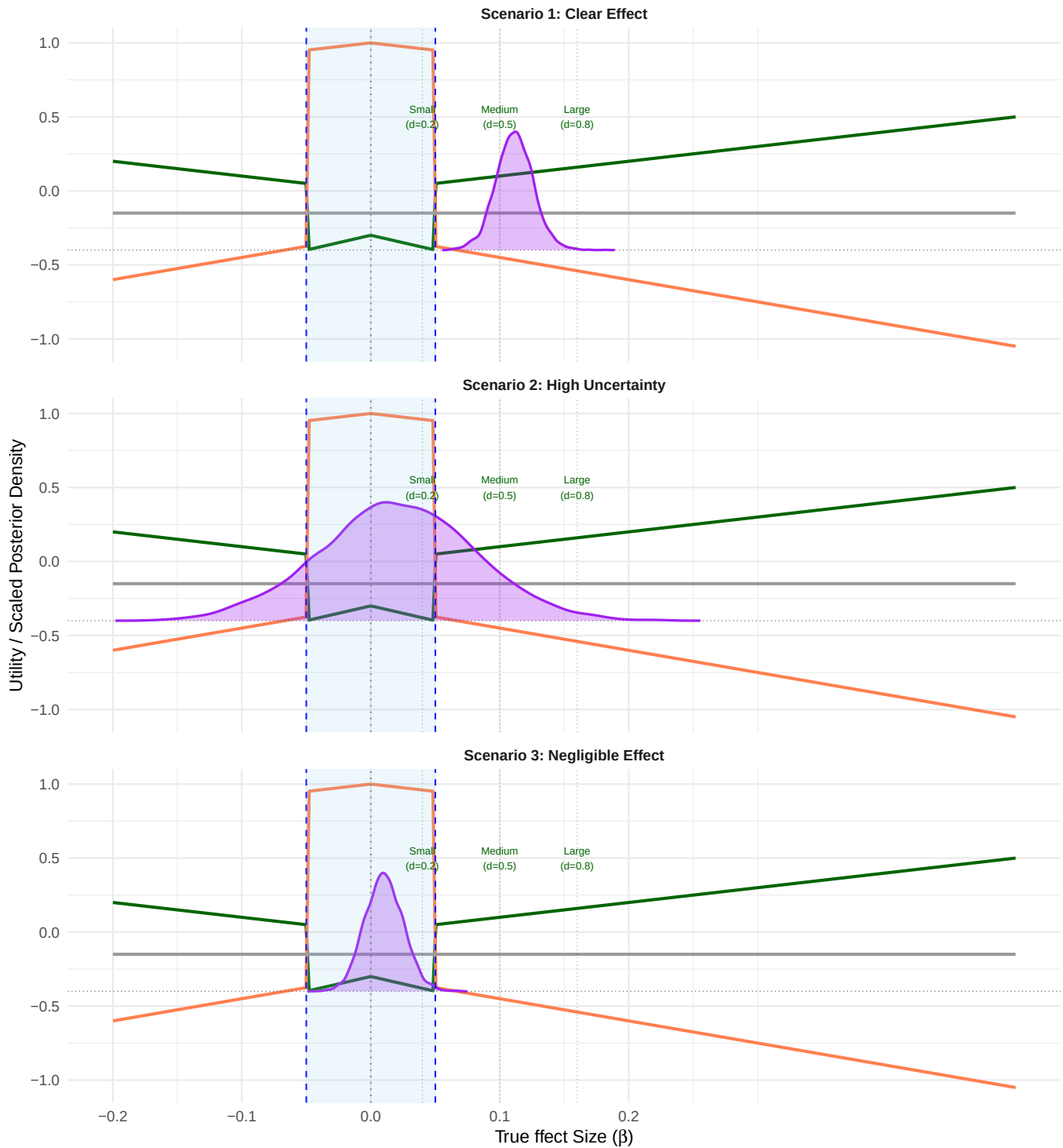
Scenario: You must decide whether to claim “Condition B is meaningfully slower” or “Condition B is essentially equivalent to A”.

Utility function (simplified for illustration):

How Posterior Uncertainty Determines Optimal Decision

Purple = posterior distribution; Green line utility grows with effect size (showing Small/Medium/Large benchmarks)

Decision — Claim meaningful — Claim negligible — Undecided



What these three scenarios show:

1. **Scenario 1 (Clear Effect):** Posterior mostly outside ROPE

- Most purple area falls where green line is highest
 - **Optimal decision:** Claim meaningful effect
 - High expected utility for “claim meaningful”
2. **Scenario 2 (High Uncertainty):** Posterior spans ROPE boundaries
 - Purple area spreads across both inside and outside ROPE
 - Risk of being wrong is high for either claim
 - **Optimal decision:** Undecided (collect more data)
 - Expected utility of claims reduced by uncertainty; -0.15 cost of more data is worth it
 3. **Scenario 3 (Negligible Effect):** Posterior concentrated inside ROPE near zero
 - Most purple area falls where orange line is highest
 - **Optimal decision:** Claim negligible effect
 - High expected utility for “claim negligible”

Key insight: The optimal decision emerges from integrating the utility curves (colored lines) over the posterior (purple distribution). Where your posterior mass concentrates determines which decision maximizes expected utility.

i Understanding the utility curves

1. Why different maximum values?

- **Orange line (claim_negligible):** utility = $1 - ||$ when correct
 - Maximum = 1.0 when $= 0$ exactly (finding true null is maximally valuable)
 - Decreases as $=$ moves toward ROPE boundary
 - Reflects: “Confirming no effect is valuable, but less so as effect approaches boundary”
- **Green line (claim_meaningful):** utility = $||$ when correct
 - Utility = effect size itself (no upper bound)
 - At $= 0.12$, utility = 0.12; at $= 0.25$, utility = 0.25
 - Reflects: “Larger effects are more valuable to discover (stronger evidence, bigger impact)”
- This asymmetry is a **modeling choice** - you could define utilities differently based on your field’s values

2. Inside ROPE ($|| < 0.05$): “Claim negligible” has highest utility (orange line near +1 when 0)

3. Outside ROPE ($|| > 0.05$): “Claim meaningful” has highest utility (green line increases with $||$)

4. At ROPE boundaries ($|| = 0.05$): Both claim decisions have negative utility

- At $= \pm 0.05$ exactly:
 - claim_meaningful utility: $-2(0.05) - 0.3 = \mathbf{-0.40}$ (wrong decision, heavy penalty)
 - claim_negligible utility: $-1.5(0.05) - 0.3 = \mathbf{-0.375}$ (wrong decision, moderate penalty)
 - undecided utility: $\mathbf{-0.15}$ (cost of more data)
- Both wrong decisions incur the -0.3 penalty (conceptual cost of “being wrong” - misleading literature, wasted resources, wrong theory) plus proportional loss

5. When “undecided” is optimal:

- **KEY:** This plot shows utility IF the true $=$ is known, but we don’t know $=$ - we have uncertainty (posterior)
- “Undecided” is optimal when **posterior uncertainty spans the boundaries**
- If your 95% HDI covers, say, $[-0.02, 0.08]$:

- Some posterior mass inside ROPE (favors “negligible”)
- Some posterior mass outside ROPE (favors “meaningful”)
- **Expected utility** of either claim is lowered by risk of being wrong
- **Expected utility** of “undecided” (-0.15) can beat both risky claims
- The grey line isn’t “highest” at any single value, but wins when averaging across uncertain

6. Asymmetric losses: False positive penalty stronger than false negative

- **Green line inside ROPE:** Drops to $-2 \times | \cdot | - 0.3$ (steep slope, heavy penalty)
- **Orange line outside ROPE:** Drops to $-1.5 \times | \cdot | - 0.3$ (moderate slope, lighter penalty)
- This asymmetry means: claiming a meaningful effect when it’s negligible (false positive) is penalized more heavily

7. The ROPE boundaries (± 0.05) are where the utility functions cross zero

- Inside: orange (claim negligible) is positive, green (claim meaningful) is negative
- Outside: green (claim meaningful) is positive, orange (claim negligible) is negative
- This defines the decision boundary
- Larger true effects $\rightarrow$ higher utility for correct “meaningful” claim (green line increases)
- Effects closer to zero $\rightarrow$ higher utility for correct “negligible” claim (orange increases toward 1 as $\rightarrow 0$)
- Being wrong by a lot is worse than being wrong by a little

! ROPE Boundaries = Utility Crossover Points

Your ROPE boundaries should be set where the utilities of “claim meaningful” and “claim negligible” are equal. This is your **smallest effect size of interest (SESIOI)**.

Computing expected utilities:

	decision	expected_utility
1	claim_meaningful	0.1114245
2	undecided	-0.1500000
3	claim_negligible	-0.4671367

The optimal decision is to **claim_meaningful** (maximizes expected utility).

i From Decision Theory to ROPE

The traditional ROPE decision rule: - “If 95% HDI excludes ROPE $\rightarrow$ Reject null”

...is actually an approximation to: - “Choose the decision with maximum expected utility”

The HDI-based rule works well when:

1. Losses are approximately symmetric
2. We want to control error rates at $\sim 5\%$
3. We prefer simple rules over computing expected utilities

For asymmetric losses or complex decisions, computing expected utilities explicitly (as shown above) provides more principled decisions.

3.2.2 The Same Graph But Simpler

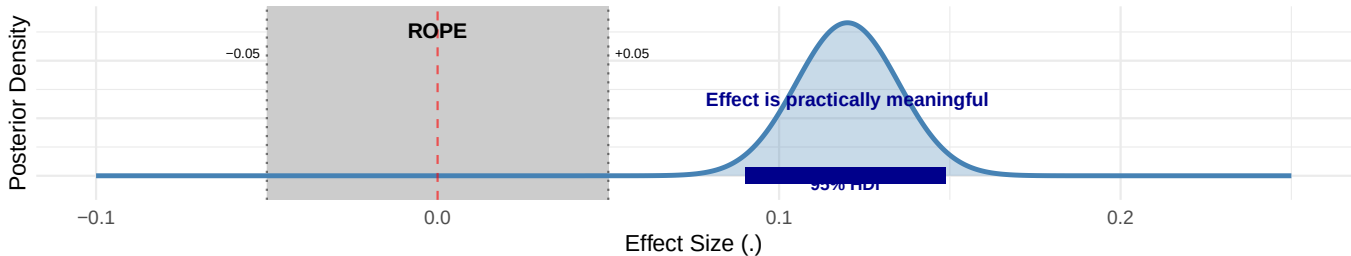
ROPE analysis has **three possible outcomes**, not just two as in NHST!

Three Possible ROPE Outcomes

ROPE boundaries: $[-0.05, +0.05]$ | Shaded gray region = ROPE | Thick blue line = 95% HDI

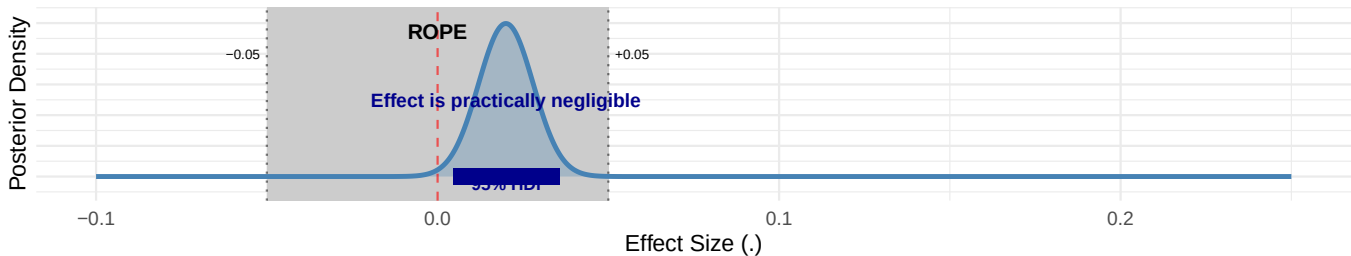
Decisive: Reject H.

HDI excludes ROPE



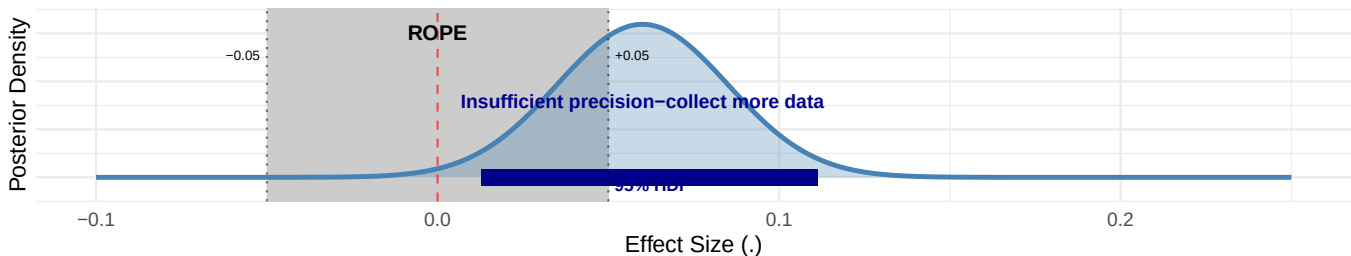
Decisive: Accept H.

HDI inside ROPE



Undecided

HDI overlaps ROPE



! Three Outcomes, Not Two!

Current practice shows:

1. **HDI excludes ROPE** → Reject H (effect is meaningful)
 - Example: 95% HDI = $[0.09, 0.15]$, ROPE = $[-0.05, 0.05]$
 - Interpretation: “Effect is decisively larger than our practical significance threshold”
2. **HDI inside ROPE** → Accept H (effect is negligible)
 - Example: 95% HDI = $[0.01, 0.03]$, ROPE = $[-0.05, 0.05]$
 - Interpretation: “Effect is decisively smaller than our practical significance threshold”
3. **HDI overlaps ROPE** → UNDECIDED (insufficient precision)
 - Example: 95% HDI = $[0.03, 0.09]$, ROPE = $[-0.05, 0.05]$
 - Interpretation: “We cannot make a clear decision—some credible values are meaningful, others aren’t”

- **This is not a failure!** It's honest reporting of uncertainty

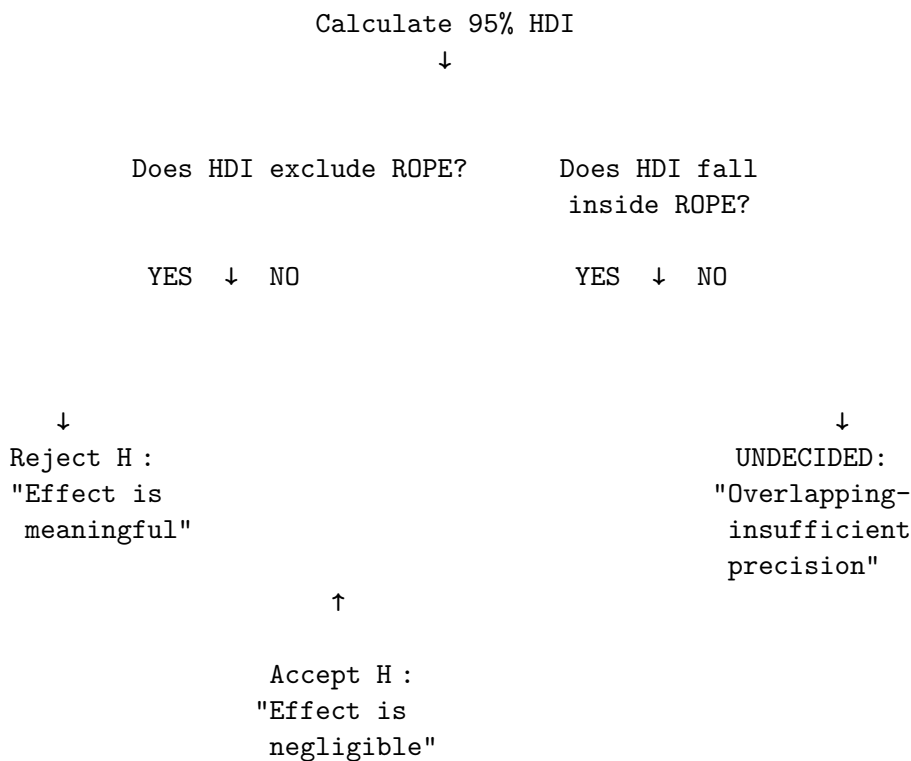
i "Undecided" Is a Feature, Not a Bug

From Kruschke (2015, p. 338):

"Be clear that any discrete decision about rejecting or accepting a null value does not exhaustively capture our knowledge about the parameter value. Our knowledge about the parameter value is described by the **full posterior distribution**."

When HDI overlaps ROPE: - You have learned something: The effect might or might not be meaningful - Your options: 1. Collect more data to narrow the posterior 2. Accept the uncertainty and make a practical decision based on other factors 3. Use a less conservative threshold (e.g., 89% HDI instead of 95%) - **Don't** force a binary decision when the data don't support one!

3.2.3 ROPE Decision Flowchart



i Manual ROPE Calculation (HDI-Based Approximation)

ROPE is simpler than computing utilities explicitly and works well for most research questions.

In practice, we usually use the HDI-based ROPE approximation rather than computing expected utilities explicitly. This is computationally simpler and works well when: - Losses are approximately symmetric (false positive false negative) - We want standard 95% decision threshold - We prefer simple rules over custom utility functions

Now let's use the traditional ROPE decision rules (which approximate the decision-theoretic framework):

```
# A tibble: 5 x 2
  Measure      Value
  <chr>      <chr>
1 ROPE boundaries [-0.05, 0.05]
2 95% HDI      [0.084, 0.142]
3 % Below ROPE 0.0%
4 % Inside ROPE 0.0%
5 % Above ROPE 100.0%
[1] **REJECT equivalence**: Effect is practically meaningful (positive) - 95% HDI entirely above
```

Concepts So Far

Before moving on, you should understand: - ROPE boundaries represent the smallest effect you care about - Three possible decisions: Accept H₀, Accept H_a, or remain undecided - Decision based on whether 95% HDI overlaps, excludes, or is inside ROPE - This approximates choosing the decision with maximum expected utility

Decision Rule = Maximizing Expected Utility

What just happened in decision-theoretic terms:

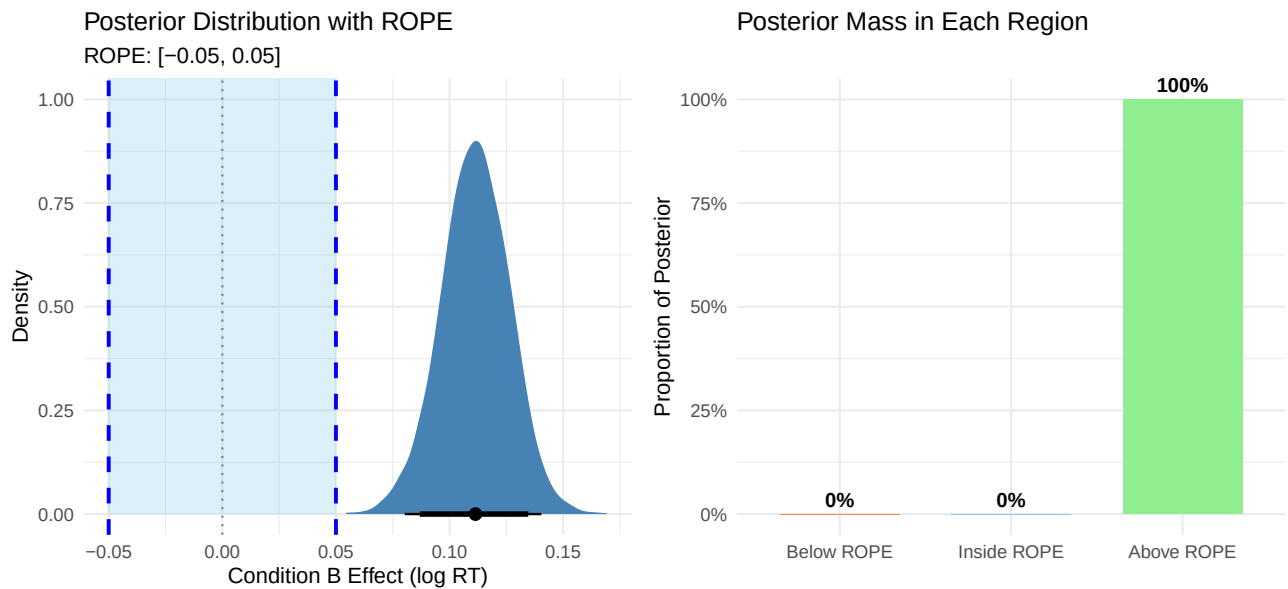
1. **We computed posterior probabilities:** $P(\text{in each region} \mid \text{Data})$
2. **We applied a decision rule:** Based on where 95% HDI falls
3. **This approximates:** Choosing decision with maximum expected utility

The connection: - When 95% HDI > ROPE: $E[U(\text{"meaningful"})] > E[U(\text{"negligible"})]$ with high confidence - When 95% HDI < ROPE: $E[U(\text{"negligible"})] > E[U(\text{"meaningful"})]$ with high confidence - When overlapping: Expected utilities too close to call → “undecided” optimal

The 95% threshold encodes a loss function where: - We’re willing to accept 5% risk of wrong decision - Losses are approximately symmetric for false positives vs. false negatives - If your losses are **asymmetric** (e.g., false positives much worse), you should: - Use stricter threshold (e.g., 99% HDI), OR - Compute expected utilities explicitly (as shown in previous section)

3.2.4 Visualizing ROPE

Manual ROPE Analysis



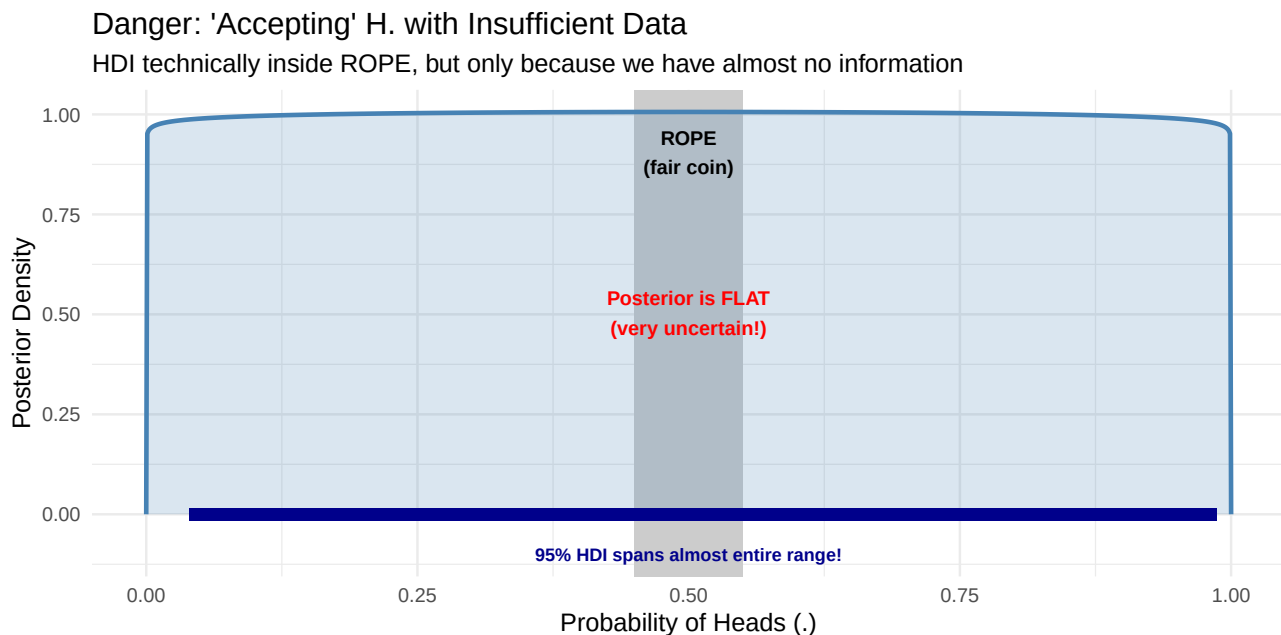
3.3 Warning

⚠ The Precision Problem

From Kruschke (2015, Section 12.2.1.1): Even with very little data, you can “accept” the null if the posterior is flat enough!

This is one of the most overlooked caveats about ROPE analysis.

Dangerous Example: Accepting H_0 with Poor Data



```
# A tibble: 3 x 2
  Measure      Value
  <chr>        <chr>
1 Data        1 head out of 2 flips
2 95% HDI for [0.039, 0.986]
3 HDI Width    0.947
```

Kruschke's solution (p. 348):

"High precision demands a large sample size... But when we are trying to accept a specific value of θ , it seems logically appropriate that we should have a reasonably precise estimate indicating that specific value."

3.3.1 Three Checks Before Trusting ROPE

Before trusting ROPE conclusions when accepting H_0 , verify precision with these three checks:

1. HDI Width: Is your estimate precise enough?

- **Rule:** HDI width should be $<$ half the ROPE width
- **Rationale:** If HDI is wide relative to ROPE, you lack precision to confidently say effect is negligible
- **Example:** ROPE = $[-0.05, 0.05]$ (width 0.10) $\rightarrow$ HDI width should be < 0.05
- **If fail:** Collect more data before claiming negligible effect

2. Effective Sample Size (ESS): Is your posterior reliable?

- **Rule:** ESS bulk > 1000 AND ESS tail > 1000
- **Rationale:** Low ESS means MCMC chains haven't converged well - posterior estimates unreliable
- **ESS bulk:** Measures sampling efficiency for central posterior
- **ESS tail:** Measures sampling efficiency for HDI boundaries (critical for ROPE!)
- **If fail:** Increase MCMC iterations, check convergence diagnostics ($\hat{R}$), consider reparameterization

3. Posterior SD: Is uncertainty manageable relative to ROPE?

- **Rule:** Posterior SD < half the ROPE width (SD/ROPE ratio < 0.5)
- **Rationale:** Large SD means high uncertainty - even if HDI is inside ROPE, estimate may be unstable
- **Example:** ROPE width = 0.10 → Posterior SD should be < 0.05
- **If fail:** Either collect more data OR accept that you're at the limits of what your design can resolve

```
# A tibble: 8 x 2
  Check                               Value
  <chr>                               <chr>
1 "1. HDI Width"                     0.0576
2 "  Status"                         FAIL (> 0.05) - Need more data
3 "2. Effective Sample Size (bulk)" 2387
4 "  Effective Sample Size (tail)" 2584
5 "  Status"                         PASS (> 1000)
6 "3. Posterior SD"                 0.0146
7 "  SD/ROPE Ratio"                 0.15
8 "  Status"                         PASS (< half ROPE)
```

Note: The HDI width check flagged a warning (0.0576 > 0.05 threshold). This indicates moderate precision—sufficient for rejecting H₀ (since HDI excludes ROPE), but we'd want narrower estimates if claiming equivalence. With N=50 subjects, the HDI width would likely pass this check.

3.3.2 Scenario Examples

Scenario 1: Wide HDI Inside ROPE

- 95% HDI: [0.01, 0.04] (width = 0.03)
- ROPE: [-0.05, 0.05]
- **WRONG:** "Effect is negligible"
- **RIGHT:** "HDI inside ROPE, but width (0.03) suggests moderate uncertainty. More data needed."

Scenario 2: Narrow HDI Inside ROPE

- 95% HDI: [0.015, 0.025] (width = 0.01)
- ROPE: [-0.05, 0.05]
- **RIGHT:** "Effect is negligible. HDI is narrow (0.01) and falls well within ROPE."

Scenario 3: HDI Overlaps ROPE Boundary

- 95% HDI: [0.03, 0.08]
- ROPE: [-0.05, 0.05]
- **RIGHT:** "Undecided. Some credible values fall within ROPE, others exceed it. More data needed."

! Rules of Thumb for Precision

Before using ROPE to **accept H₀** (claim negligible effect), verify:

For RT effects (log-scale):

- HDI width < 0.05 log-units
- Posterior SD < 0.025 (half the typical ROPE width)
 - *Why?* Larger SD means too much uncertainty about whether the effect is truly negligible.

For accuracy (probability scale):

- HDI width < 0.10
- Posterior SD < 0.05

For all analyses:

- ESS (bulk) > 1000
- ESS (tail) > 1000
- Convergence diagnostics passed ($R < 1.01$)

If these checks fail:

- Don't claim effect is negligible
- Report "insufficient precision" or "undecided"
- Consider collecting more data

💡 When You Can Trust ROPE to Accept H

Strong evidence for negligible effect requires:

1. **Narrow posterior:** HDI width $<$ half ROPE width
2. **Central location:** HDI midpoint near zero
3. **Good sampling:** ESS > 1000 , $R < 1.01$
4. **Model fit:** Posterior predictive checks pass

Example of strong evidence:

Effect: = 0.02 (95% HDI: [0.01, 0.03])

ROPE: [-0.05, 0.05]

HDI width: 0.02 (narrow)

ESS: 2500 (adequate)

→ Can confidently claim negligible effect

Example of weak evidence:

Effect: = 0.02 (95% HDI: [-0.01, 0.05])

ROPE: [-0.05, 0.05]

HDI width: 0.06 (too wide)

ESS: 800 (marginal)

→ Cannot confidently claim negligible effect

3.4 Using bayestestR Package

The `bayestestR` package provides convenient functions for ROPE analysis.

```
# Proportion of samples inside the ROPE [-0.05, 0.05]:
```

```
Parameter | Inside ROPE
-----
Intercept |      0.00 %
conditionB |      0.00 %
```

3.4.1 Interpretation

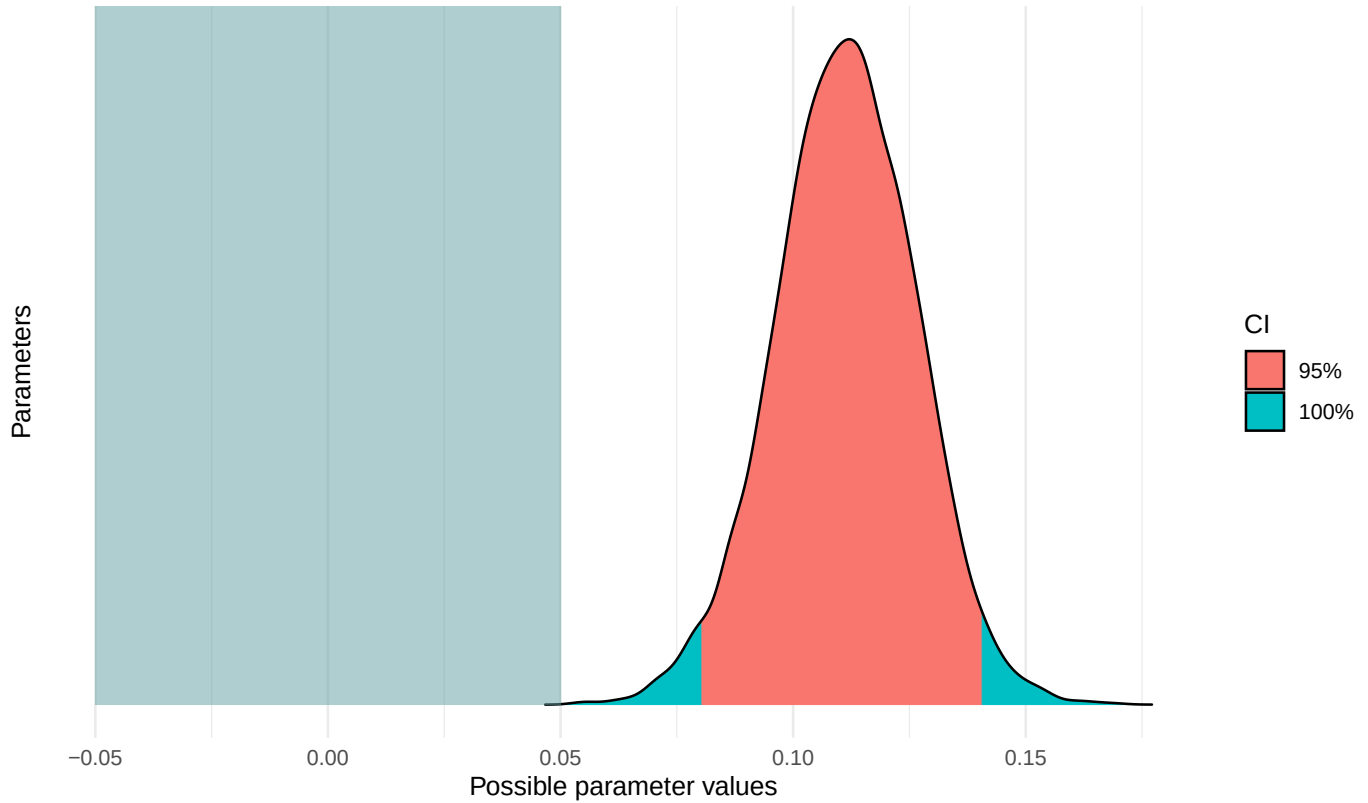
The output shows:

- **% in ROPE:** Proportion of 95% HDI inside ROPE

- Interpretation automatically provided

3.4.2 Visualizing with bayestestR

ROPE Analysis using bayestestR



3.4.3 Equivalence Test

The `equivalence_test()` function provides an integrated view:

```
# Test for Practical Equivalence
```

```
ROPE: [-0.05 0.05]
```

Parameter	H0	inside ROPE	95% HDI
Intercept	Rejected	0.00 %	[5.90, 6.05]
conditionB	Rejected	0.00 %	[0.08, 0.14]

This shows for each parameter:

- **% in ROPE:** How much of the posterior falls in ROPE
- **Decision:** Accepted/Rejected/Undecided

i Comparison: Manual vs bayestestR

Both approaches should give the same results:

```
# A tibble: 2 x 3
  Source          `"% in ROPE"` Decision
  <chr>              <dbl> <chr>
1 Manual Calculation      0 Rejected
2 bayestestR Package      0 Rejected
```

What Just Happened?

Both approaches give identical results — bayestestR simply automates the manual calculations we did earlier. The benefit: - Less code to write - Automatic formatting - Built-in visualization - Consistent interface across analyses

When to Use Manual vs Package

Use manual calculation when:

- You want to understand the mechanics
- You need custom ROPE boundaries for different parameters
- You're teaching/explaining the concept

Use bayestestR when:

- You want quick, standardized results
- You're analyzing many models
- You want built-in visualization

i Prior Sensitivity Analysis for ROPE Decisions

When Priors Matter

From Kruschke (2015, Section 12.2.1): ROPE conclusions can change with different priors.

Always check sensitivity, especially when:

- Sample size is small ($n < 30$ per group)
- HDI barely touches ROPE boundary
- You're accepting H (claiming negligible effect)

3.4.4 Demonstration: Same Data, Different Priors, Different ROPE Conclusions

	Prior	CI	ROPE_low	ROPE_high	ROPE_Percentage
1 Weakly Informative	$N(0, 0.5)$	0.95	-0.05	0.05	0.23894737
2 Skeptical	$N(0, 0.10)$	0.95	-0.05	0.05	0.09894737
3 Diffuse	$N(0, 2)$	0.95	-0.05	0.05	0.06157895

Decision

1 Undecided

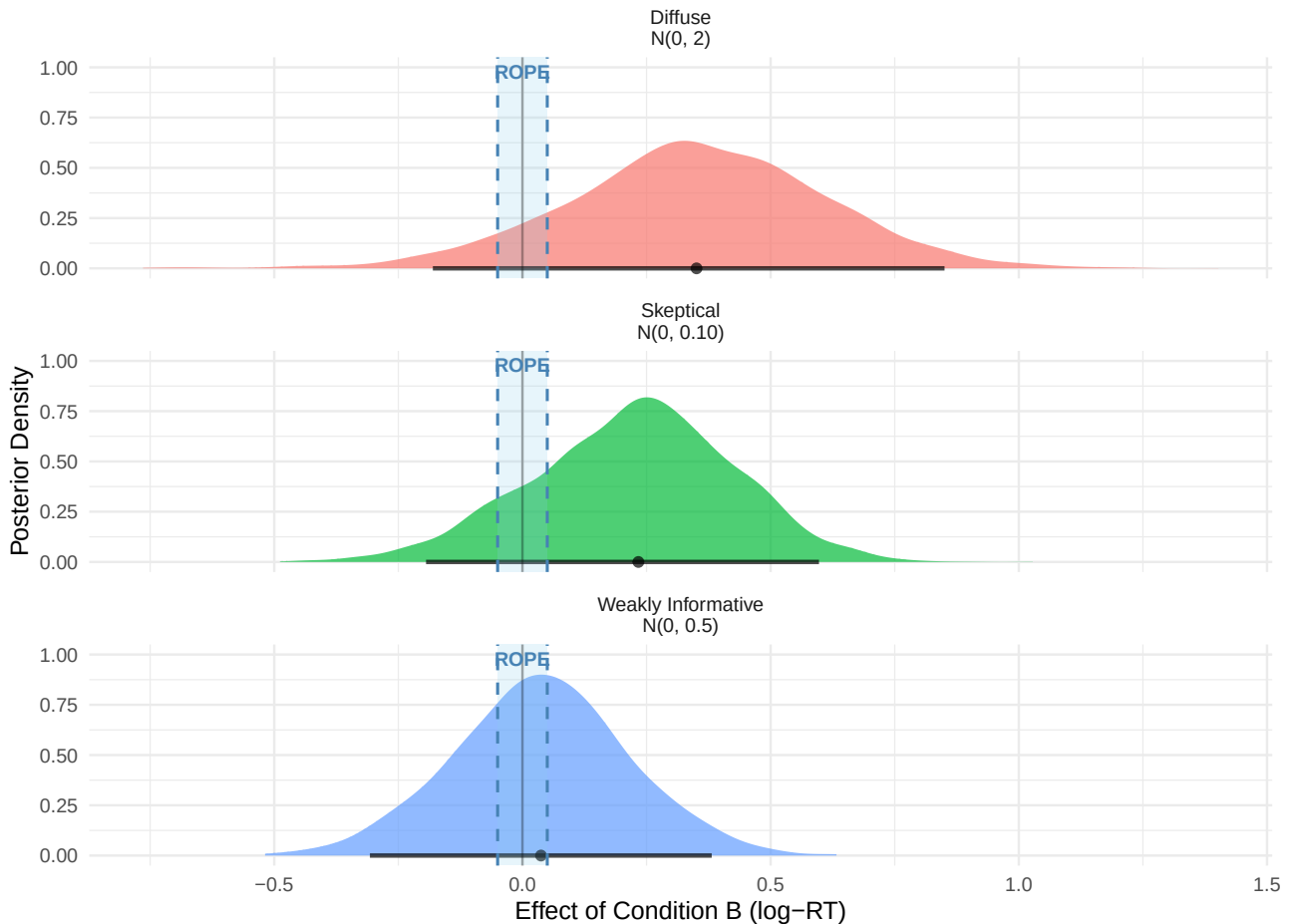
2 Undecided

3 Undecided

3.4.5 Visualize Prior Sensitivity

ROPE Sensitivity to Prior Choice

Same data (N=15 subjects), different priors → How much do conclusions change?



3.4.6 When ROPE Conclusions Hold vs. Change

ROPE Conclusions Hold When:

Large sample size ($n > 30$ per group)

- Data overwhelms prior
- Posterior similar regardless of prior choice

Effect clearly outside or inside ROPE

- HDI far from ROPE boundaries
- All reasonable priors give same conclusion

Narrow posterior

- HDI width $\ll$ ROPE width
- High precision reduces prior influence

ROPE Conclusions Are Sensitive When:

Small sample size ($n < 20$ per group) - Prior has substantial influence - Different priors can change conclusion

HDI barely touches ROPE boundary - Effect 0.05 (right at threshold) - Small

prior differences flip decision

Wide posterior - HDI width ROPE width - Uncertainty dominates

Example of sensitivity:

Data: N=15 subjects, effect 0.04 log-units

ROPE: [-0.05, 0.05]

Result:

- Skeptical prior: 100% in ROPE (accept H)
- Diffuse prior: 60% in ROPE (undecided)

→ Conclusion sensitive!

3.4.7 What To Do If Conclusions Are Sensitive

3.4.7.1 Option 1: Report Multiple Analyses (Recommended)

Report all results from different prior specifications:

3.4.7.2 Option 2: Justify Prior More Carefully

If conclusions are sensitive, invest more effort in prior justification:

- Pilot data
- Expert elicitation
- Previous literature
- Prior predictive checks (Module 02)

Report: “We used prior [X] because [justification]. Given potential sensitivity with small N, we verified conclusions hold with more diffuse prior.”

3.4.7.3 Option 3: Collect More Data

If:

- Conclusions sensitive AND
- You cannot justify prior AND
- Decision matters

→ Collect more data before making decision

This is NOT Optional Stopping / P-Hacking!

In NHST: Looking at data, then deciding to collect more is a questionable research practice (QRP) - it inflates Type I error because your stopping rule affects the p-value.

In Bayesian analysis: You CAN look at the data and decide to collect more without invalidating inference!

Why? Bayesian inference follows the **likelihood principle**:

- The posterior depends **ONLY** on the data and prior
- NOT on your stopping intentions or whether you peeked
- Looking at N=15, seeing inconclusive results, and collecting to N=60 gives the **SAME** posterior as pre-planning N=60

The difference:

- **NHST QRP:** “Keep collecting until $p < 0.05$ ” (invalidates test)
- **Bayesian valid:** “Keep collecting until HDI is narrow enough for confident decision” (perfectly legitimate)

Caveat: You still shouldn’t change your ROPE boundaries or priors after seeing results!

Planning how much more data you need:

Use this power analysis to estimate target sample size for adequate precision:

```
# Rule of thumb: HDI width inversely proportional to sqrt(n)
current_n <- 8          # Current sample size
current_hdi_width <- 0.10 # Current HDI width from preliminary analysis
desired_hdi_width <- 0.05 # Want HDI < half ROPE width for confident conclusions

# Calculate target N needed
target_n <- current_n * (current_hdi_width / desired_hdi_width)^2

tibble(
  Measure = c("Current N", "Current HDI width", "Desired HDI width", "Target N needed", "Additional N needed"),
  Value = c(current_n, current_hdi_width, desired_hdi_width, round(target_n), round(target_n - current_n))
)
```

Interpretation: To reduce HDI width from 0.10 to 0.05, you need to increase N by a factor of 4 (halving width requires quadrupling sample size).

3.4.7.4 Option 4: Be Transparent About Uncertainty

If data collection isn't possible:

```
"Given our sample size (N=15), ROPE conclusions were sensitive to prior
specification. With weakly informative priors, X% of posterior fell in
ROPE, suggesting [interpretation]. With more skeptical priors, Y% fell
in ROPE, suggesting [alternative interpretation]. We recommend treating
this finding as preliminary pending replication with larger sample."
```

3.4.8 Worked Example: Prior Sensitivity Reporting

A tibble: 3 x 4

Prior	Estimate	`95% HDI`	`% in ROPE`
<chr>	<dbl>	<chr>	<dbl>
1 1. Weakly informative: N(0, 0.5)	0.037	[-0.307, 0.381]	0.2
2 2. Skeptical: N(0, 0.10)	0.223	[-0.194, 0.597]	0.1
3 3. Diffuse: N(0, 2)	0.35	[-0.180, 0.850]	0.1

Conclusion: ROPE decision holds across prior specifications. All three priors yield: Undecided

Main Point

Prior sensitivity is not a bug—it's a feature!

It tells you when your data are: - **Strong enough** to overcome prior beliefs → Confident conclusions - **Too weak** to overcome prior beliefs (sensitive) → Need more data or transparency

NHST doesn't have this diagnostic—it can give you “ $p < 0.05$ ” even with barely any data, hiding the fact that different assumptions would give different answers.

Bayesian analysis makes sensitivity **explicit and quantifiable**.

4 Understanding Why ROPE Works: Decision Theory

From Practice to Theory

Now that you’ve seen ROPE in action, let’s understand **why** it works. This section explains the decision-theoretic foundations. While not required for using ROPE, it helps you understand what ROPE boundaries mean and when to adjust them.

4.1 The Decision-Theoretic Framework

4.1.1 Making Decisions Under Uncertainty

When we analyze data, we’re ultimately making **decisions**:

- Should we claim an effect exists and is meaningful?
- Should we claim it’s negligible?
- Should we collect more data?

Each decision has **consequences** (utilities or losses):

- **Claiming meaningful effect when it’s negligible**: False positive (wasted resources, misleading literature)
- **Claiming negligible effect when it’s meaningful**: False negative (missed opportunity, wrong theory)
- **Remaining undecided**: Cost of additional data collection

This is **Bayesian decision analysis** (Gelman et al., 2013; see [Stan User’s Guide](#)):

Theoretical Framework (Four Components):

1. **Define outcomes and decisions**
 - Outcomes: Possible values of the effect ()
 - Decisions: {Accept H_0 (negligible), Accept H_1 (meaningful), Remain undecided}
2. **Define probability distribution**
 - Our posterior distribution $p(\theta \mid \text{Data})$
 - Quantifies uncertainty about the effect size
3. **Define utility (or loss) function**
 - $U(\theta, \text{decision})$: What’s the cost/benefit of each decision for each possible true effect?
 - This is where **domain knowledge** enters!
4. **Choose optimal decision**
 - Maximize expected utility (minimize expected loss)
 - $d^* = \arg \max_d E[U(\theta, d) \mid \text{Data}]$
 - (Translation: “Choose decision d that maximizes expected utility, averaging over posterior uncertainty in θ ”)

4.1.2 ROPE as Decision Analysis

ROPE implements this framework!

When you set ROPE boundaries at $[-0.05, +0.05]$, you’re defining a **loss function**:

`Loss( , decision):`

If decide "meaningful effect":

- Low loss if $|| > 0.05$ (correct decision)
- HIGH LOSS if $|| < 0.05$ (false positive)

If decide "negligible effect":

- Low loss if $|| < 0.05$ (correct decision)
- HIGH LOSS if $|| > 0.05$ (false negative)

The **ROPE decision rule** minimizes expected loss:

- If 95% HDI excludes ROPE → High expected utility for “meaningful effect”
- If 95% HDI inside ROPE → High expected utility for “negligible effect”
- If overlapping → Insufficient evidence, collecting more data may have higher utility

! Why This Matters

ROPE is not arbitrary! It’s the Bayesian optimal decision given:

1. Your posterior beliefs (from the model)
2. Your utility function (encoded in ROPE boundaries)
3. Your decision threshold (e.g., 95% credibility)

Changing ROPE boundaries = changing your utility function = changing what you consider “meaningful enough to matter”.

5 Effect Estimation with emmeans

i Moving Beyond Single Comparisons

So far we’ve tested practical significance for a single effect (Condition B vs A). But what if you have **three or more groups**? You need:

- All pairwise comparisons (A vs B, A vs C, B vs C)
- ROPE analysis for each comparison
- Adjustment for multiple comparisons (optional)

This is where **emmeans** and **marginaleffects** come in.

5.1 Why emmeans?

When you have **factorial designs**, you often want to:

- Compare all pairwise combinations
- Estimate marginal means (averaged over random effects)
- Get automatic adjustment for multiple comparisons
- Use familiar syntax from frequentist stats

emmeans (estimated marginal means) provides all of this and works directly with brms!

5.2 Example: Three-Condition Design

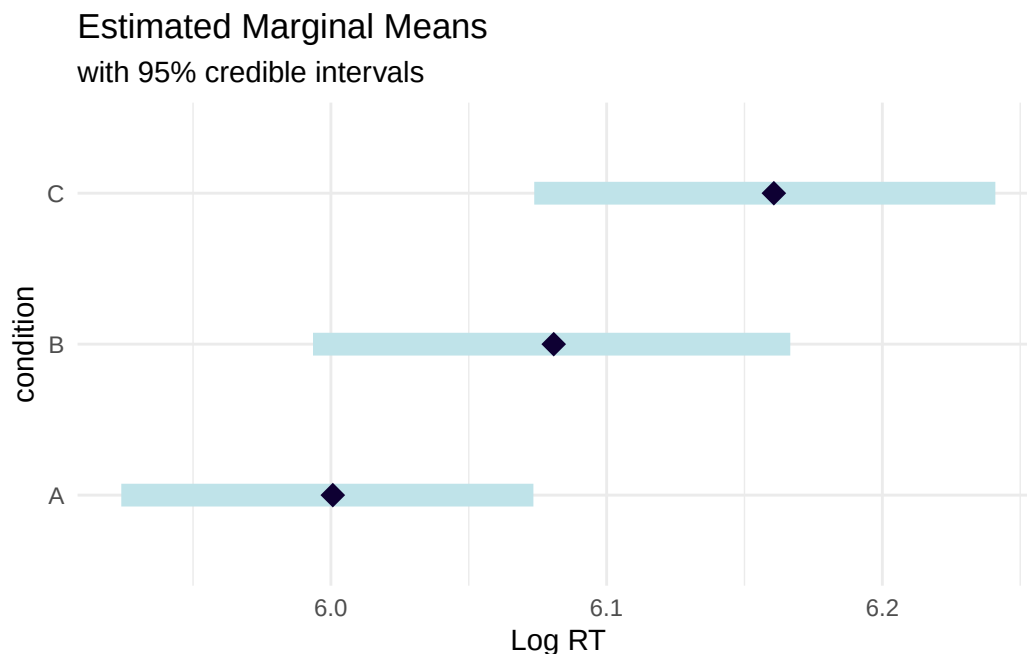
Let’s extend our example to three conditions:

```
# A tibble: 3 x 5
  condition mean_rt sd_rt mean_log_rt sd_log_rt
  <fct>      <dbl> <dbl>      <dbl>      <dbl>
1 A         411.  78.8         6.00       0.193
2 B         448.  99.7         6.08       0.221
3 C         483. 102.         6.16       0.218
```

5.3 Estimated Marginal Means

Table 5: Estimated Marginal Means (Log RT)

Condition	Mean	95% HPD
A	6.001	[5.924, 6.073]
B	6.081	[5.993, 6.167]
C	6.161	[6.074, 6.241]



i What are “Estimated Marginal Means”?

EMMs are model-predicted means that:

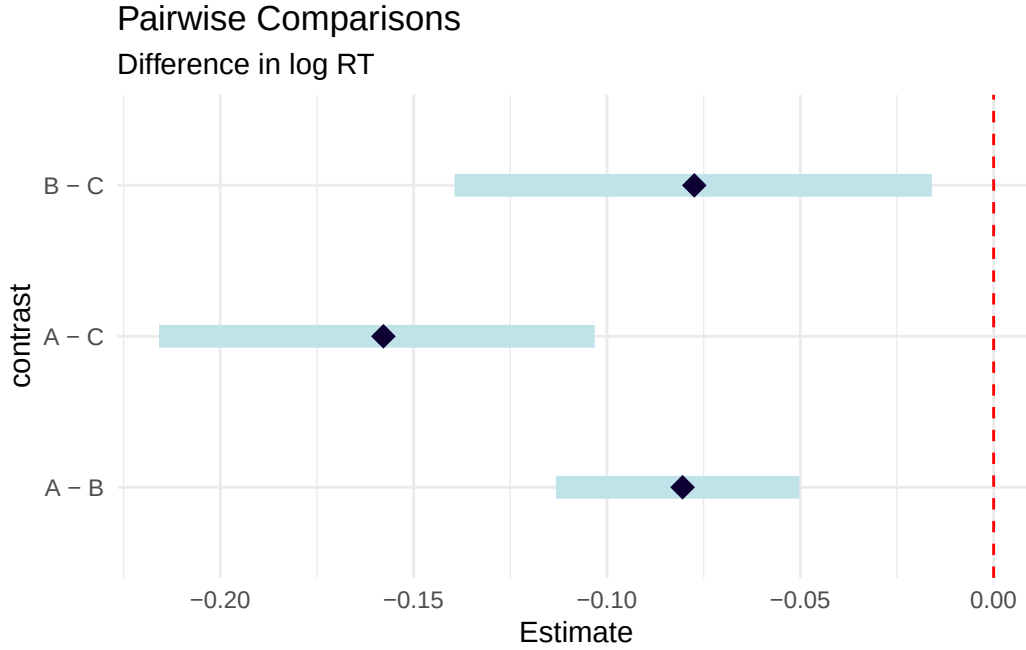
- Average over random effects (subjects, items)
- Provide population-level estimates
- Include full Bayesian uncertainty (not just point estimates!)

Think of them as “What would we expect for a typical new subject/item?”

5.4 Pairwise Comparisons

Table 6: **Pairwise Comparisons (Difference in Log RT)**

Contrast	Estimate	95% HPD
A - B	-0.080	[-0.113, -0.050]
A - C	-0.158	[-0.216, -0.103]
B - C	-0.077	[-0.139, -0.016]



5.5 ROPE Analysis on Pairwise Comparisons

Now combine emmeans with ROPE to test practical significance of each comparison:

Table 7: **ROPE Analysis Results for Pairwise Comparisons**

Comparison	Estimate	95% HDI	Result
1	NA	[-0.113, -0.050]	↓ MEANINGFUL DECREASE
2	NA	[-0.216, -0.103]	↓ MEANINGFUL DECREASE
3	NA	[-0.139, -0.016]	? UNDECIDED

5.6 Custom Contrasts

emmeans allows custom contrasts beyond pairwise comparisons:

Table 8: **Custom Contrasts**

Contrast	Estimate	95% HPD
BC_vs_A	0.119	[0.086, 0.153]
C_vs_B	0.077	[0.016, 0.139]

Table 9: **ROPE Analysis for Custom Contrasts**

Contrast	Decision
1	Meaningful positive effect
2	Undecided

Summary: emmeans

You should now understand:

- Estimated marginal means (EMMs) are population-level predictions
- `pairs()` gives all pairwise comparisons automatically
- Extract posterior samples with `as.mcmc()` to integrate with ROPE
- Works directly with brms for full Bayesian inference

When to Use emmeans

Perfect for:

- Factorial designs (2×2 , 2×3 , etc.)
- All pairwise comparisons automatically
- Familiar syntax from lsmeans/emmeans in frequentist stats
- Integration with ROPE for practical significance

Not ideal for:

- Simple two-group comparisons (just use brms coefficients)
- Complex non-linear predictions (use `marginaleffects` instead)
- Interactions with continuous predictors (`marginaleffects` better)

Effect Estimation with `marginaleffects` (Under Construction - Click to Expand)

Section Under Development

This section is currently being revised and some tables/visualizations may not display correctly. The emmeans approach (above) is fully functional and recommended for now.

6 Effect Estimation with `marginaleffects`

Modern Alternative to emmeans

While emmeans is excellent for factorial designs, **`marginaleffects`** offers:

- Unified syntax across ALL model types (not just brms)
- More flexible predictions and comparisons
- Better support for continuous predictors and interactions
- Modern tidyverse-compatible workflow

Let's see how it compares.

6.1 Why marginaeffects?

marginaeffects provides a modern, unified interface for:

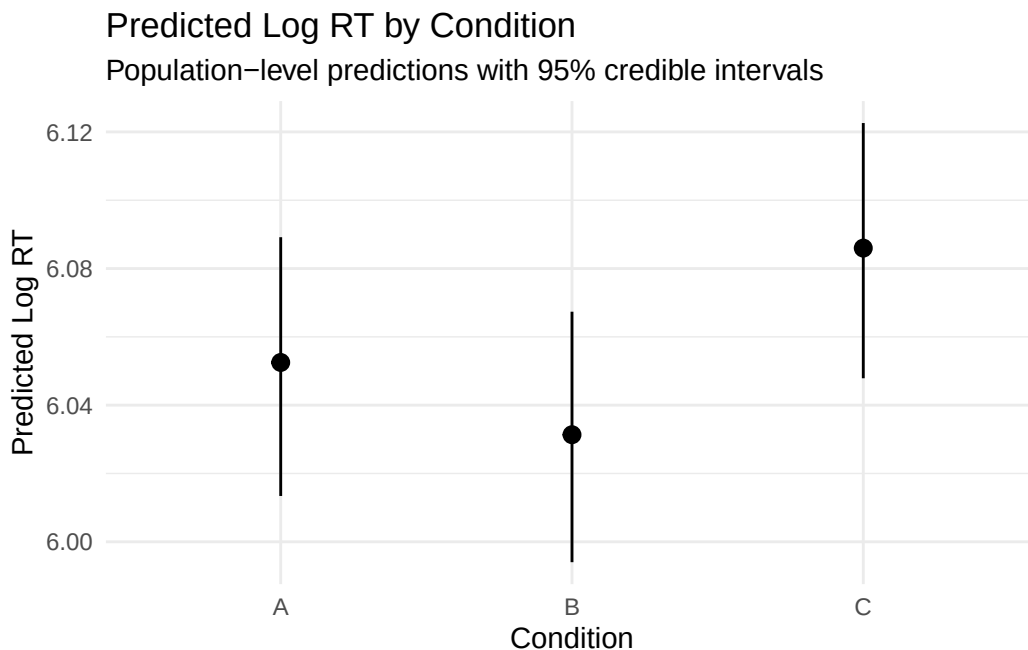
- Predictions at specific values
- Comparisons (differences, ratios, etc.)
- Slopes (derivatives) for continuous predictors
- Custom hypotheses with flexible syntax

It works with brms, rstanarm, glm, lme4, and many other models!

6.2 Visualize 95% Credible Intervals

Table 10: Predicted Log RT by Condition

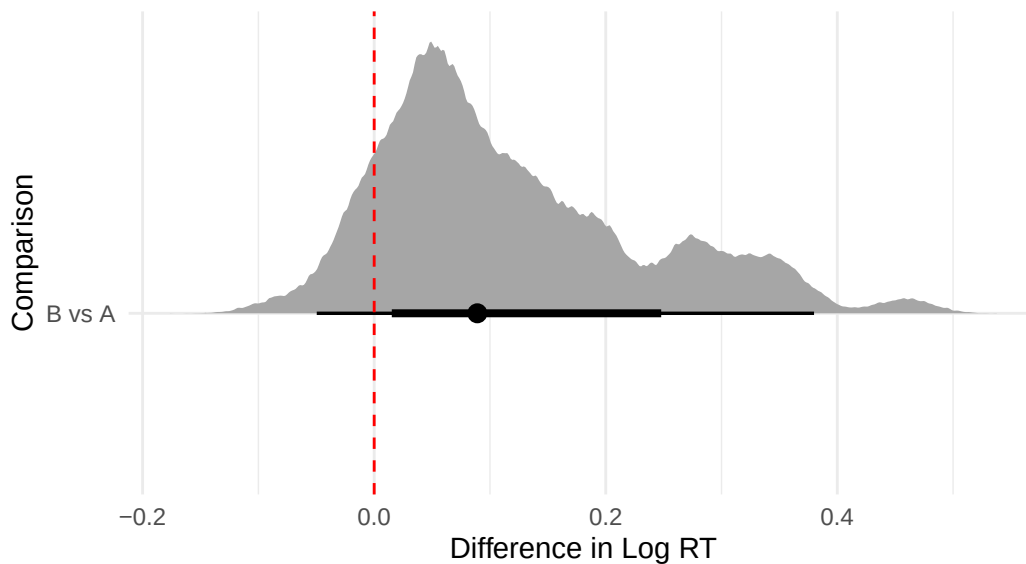
Condition	Estimate	95% CI
A	6.053	[6.013, 6.089]
B	6.031	[5.994, 6.067]
C	6.086	[6.048, 6.123]



6.3 Comparisons: How Much Do Conditions Differ?

Pairwise Comparisons

Posterior distributions of differences in log RT



6.4 Reference Grid: Custom Comparisons

Table 11: **Comparison: B vs A**

Comparison	Difference	95% CI
B vs A	-0.029	[-0.074, 0.018]

Table 12: **Custom Comparison: C vs Average(A, B)**

Measure	Value
Estimate	0.044
95% HDI	[0.003, 0.085]
% Below ROPE	0.0%
% In ROPE	61.1%
% Above ROPE	38.9%

6.4.1 Decision

Undecided - need more data

6.5 Hypotheses: Flexible Hypothesis Testing

`marginalEffects` doesn't support `hypotheses()` for Bayesian models, but we can test hypotheses using `comparisons`:

6.5.1 Interpretation Guide

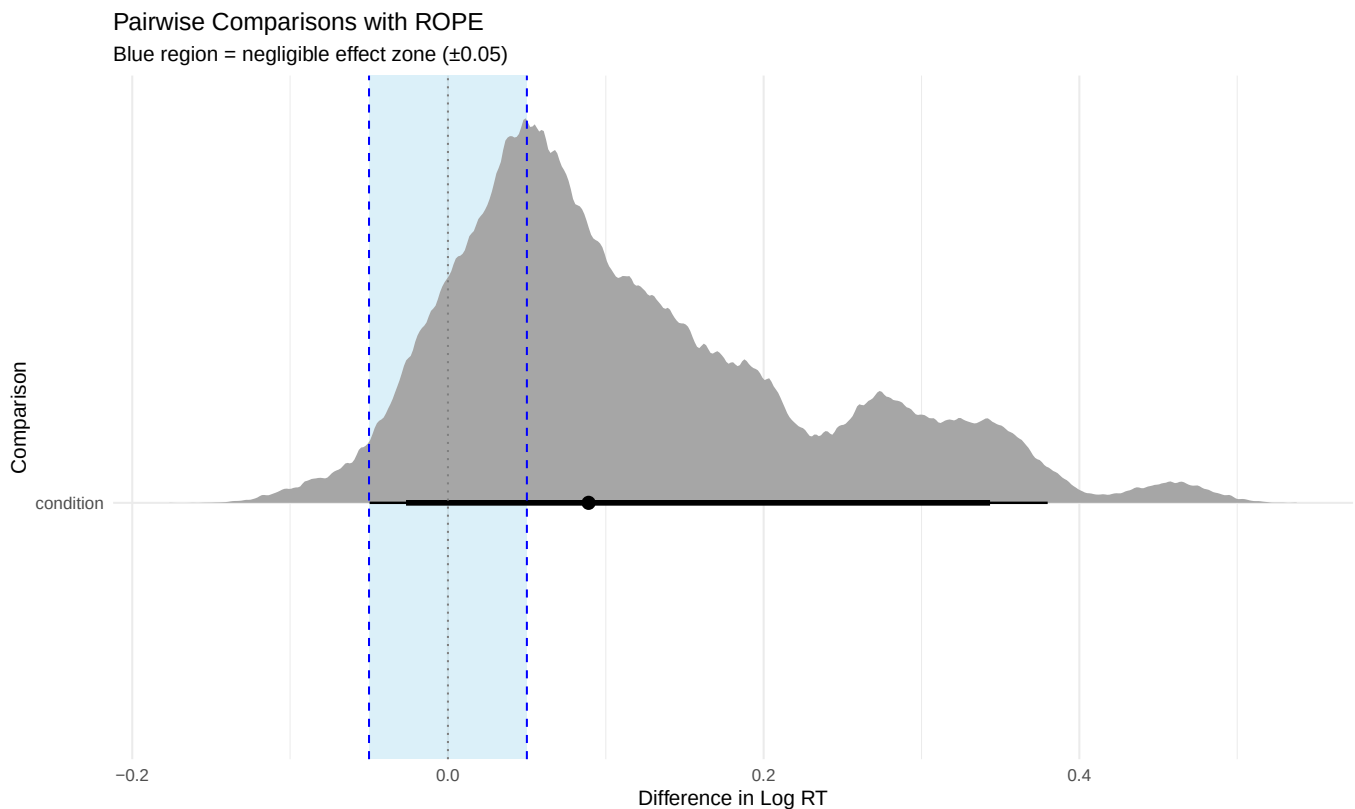
- **Estimate:** Mean difference in log-RT between conditions
- **95% CI:** Credible interval for the difference
- **P(diff > 0):** Probability first condition has higher RT than second
- **P(|diff| > 0.05):** Probability difference exceeds ± 0.05 ROPE threshold

6.6 ROPE Analysis with marginaeffects

Combine marginaeffects with ROPE:

Table 14: **ROPE Analysis Results Using marginaeffects**

Comparison	Estimate	95% HDI	% in ROPE	Result
condition	0.119	[-0.050, 0.380]	29.9%	? UNDECIDED



i Summary: marginaeffects

You should now understand:

- `predictions()` computes model predictions at specific values
- `comparisons()` computes differences between conditions
- `posterior_draws()` extracts MCMC samples for ROPE analysis
- Works with `brms`, `rstanarm`, `glm`, `lme4`, and many other packages

💡 When to Use `marginalEffects`

Perfect for:

- Any type of model (GLM, multilevel, GAM, etc.)
- Predictions at specific covariate values
- Non-linear transformations (odds ratios, percentages, etc.)
- Custom hypotheses with complex logic
- Modern, consistent syntax across models

Advantages over `emmeans`:

- More flexible predictions
- Better for continuous predictors
- Unified interface across packages
- Direct ROPE integration via `posterior_draws()`

Use `emmeans` if:

- You want traditional EMM workflow
- All pairwise comparisons with adjustment
- Familiar with `lsmeans/emmeans` syntax

7 ROPE Analysis Workflow

i Bringing It All Together

You've now learned:

- ROPE: Theory and implementation
- `emmeans`: Factorial design comparisons
- `marginalEffects`: Flexible predictions
- `tidybayes`: Visualization

Now let's integrate everything into a complete analysis workflow.

Now that we've covered all the tools (ROPE, `emmeans`, `marginalEffects`, `tidybayes`), let's see how to put them together in a complete practical significance analysis.

7.1 Step-by-Step ROPE Analysis

```
# A tibble: 2 x 2
  Measure      Value
  <chr>        <chr>
1 Posterior mean 0.111
2 95% HDI       [0.084, 0.142]
```

7.1.1 Step 2: Test Practical Significance (ROPE)

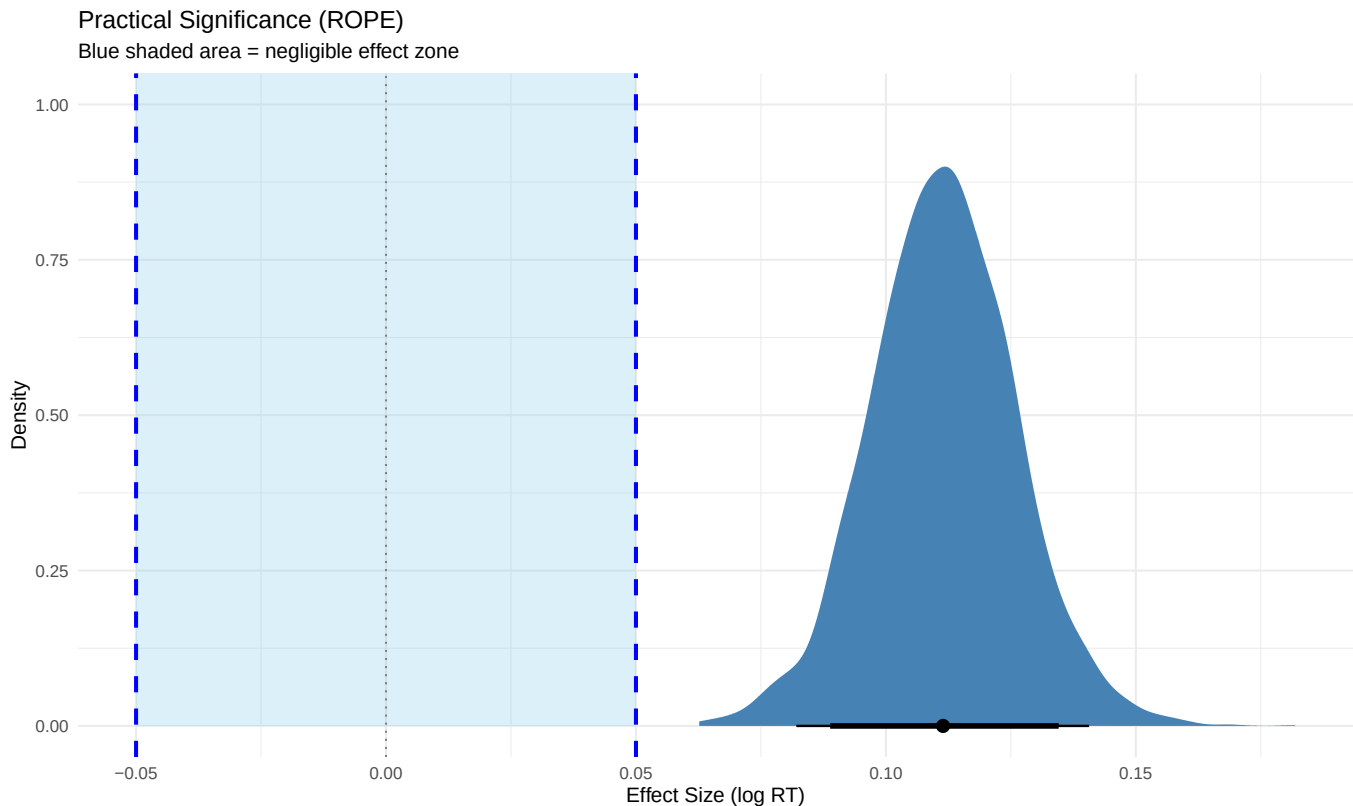
```
# A tibble: 3 x 2
  Measure      Value
  <chr>        <chr>
1 ROPE       [-0.05, +0.05] (5% RT difference)
2 % in ROPE 0.0%
```

3 Decision Effect is practically meaningful (HDI > ROPE)

7.1.2 Step 3: Integrated Conclusion

STRONG CONCLUSION: Effect is meaningful
→ Condition B differs from A
→ The difference is large enough to matter

7.2 Visualizing the Complete Picture



7.3 Complete Reporting: APA-Style Templates

! What to Always Report

From Kruschke (2015, p. 338):

“Reporting the limits of an HDI region is more informative than reporting the declaration of a reject/accept decision. By reporting the HDI and other summary information about the posterior, different readers can apply different ROPEs to decide for themselves whether a parameter is practically equivalent to a null value.”

Complete reporting includes:

1. Full posterior summary (not just inside/outside ROPE)
2. ROPE boundaries with justification
3. Effect size on multiple scales
4. Model diagnostics

7.3.1 Methods Section Template

Use this template for your Methods section when reporting ROPE analysis:

Include in Methods:

- Prior specification with rationale
- ROPE boundaries with justification
- When boundaries were set (a priori)
- How boundaries relate to original scale
- Sample size (subjects, items, observations)
- Software versions

7.3.2 Results Section Template

Use this template for your Results section:

Include in Results:

- Point estimate with HDI (on model scale)
- Posterior SD or uncertainty measure
- % of posterior in/outside ROPE
- Decision statement (reject/accept/undecided H)
- Effect size on original scale with interpretation
- Absolute magnitudes (e.g., milliseconds) where relevant
- Robustness checks (prior sensitivity, model comparison)

8 When to Use What: Decision Framework

Quick Decision Tree

What do you want to test?

- “Is the effect big enough to matter?” → Use ROPE
- “Which hypothesis is better supported?” → See Module 07 (Bayes Factors)
- “Which model fits better?” → Use LOO (Module 03)

8.1 Visual Guide: Choosing Your Tools

Your Research Question

- "Is effect meaningful?" → ROPE
- "Compare 3+ groups?"
 - Factorial design → emmeans + ROPE
 - Custom predictions → marginaleffects + ROPE
- "Which hypothesis better?" → Module 07 (Bayes Factors)

→ "Which model structure?" → Module 05 (L00)

9 Avoiding Pitfalls: Checklist

Before running analyses:

- ☐ Priors are **weakly informative** (not flat)?
- ☐ ROPE boundaries defined **before** seeing results?
- ☐ ROPE boundaries **justified** with domain knowledge?
- ☐ Hypotheses based on **theory**, not data exploration?

When interpreting results:

- ☐ Report **ROPE decision** and effect sizes?
- ☐ Report **uncertainty** (don't hide ROPE overlaps)?
- ☐ Check **prior sensitivity** (Module 04)?
- ☐ Show **posterior distributions visually**?
- ☐ Interpret on the **original scale** when possible?

10 Quick Reference

Main papers:

- **Kruschke, J. K. (2018)**. Rejecting or accepting parameter values in Bayesian estimation. *Advances in Methods and Practices in Psychological Science*, 1(2), 270-280. [The definitive ROPE paper]
- **Kruschke, J. K. (2015)**. *Doing Bayesian data analysis* (2nd ed.). Academic Press. [Chapters 11-12]
- **Makowski, D., Ben-Shachar, M. S., & Lüdtke, D. (2019)**. bayestestR: Describing effects and their uncertainty. *Journal of Open Source Software*, 4(40), 1541.
- **Lakens, D., Scheel, A. M., & Isager, P. M. (2018)**. Equivalence testing for psychological research. *Advances in Methods and Practices in Psychological Science*, 1(2), 259-269.

10.1 Session Info

R version 4.5.2 (2025-10-31)

Platform: x86_64-pc-linux-gnu

Running under: Ubuntu 24.04.3 LTS

Matrix products: default

BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3

LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.26.so; LAPACK version 3.12.0

locale:

[1] LC_CTYPE=en_US.UTF-8	LC_NUMERIC=C
[3] LC_TIME=en_US.UTF-8	LC_COLLATE=en_US.UTF-8
[5] LC_MONETARY=en_US.UTF-8	LC_MESSAGES=en_US.UTF-8
[7] LC_PAPER=en_US.UTF-8	LC_NAME=C

```
[9] LC_ADDRESS=C          LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
```

time zone: Etc/UTC

tzcode source: system (glibc)

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

other attached packages:

```
[1] knitr_1.50          patchwork_1.3.2      bayestestR_0.17.0
[4] tidybayes_3.0.7     posterior_1.6.1.9000 bayesplot_1.15.0
[7] lubridate_1.9.4     forcats_1.0.1        stringr_1.5.2
[10] dplyr_1.1.4         purrr_1.1.0          readr_2.1.5
[13] tidyr_1.3.2         tibble_3.3.0         ggplot2_4.0.1
[16] tidyverse_2.0.0     brms_2.23.0          Rcpp_1.1.0
```

loaded via a namespace (and not attached):

```
[1] svUnit_1.0.8        tidymodels_1.2.1     farver_2.1.2
[4] loo_2.8.0           S7_0.2.0             fastmap_1.2.0
[7] tensorA_0.36.2.1    digest_0.6.37        timechange_0.3.0
[10] estimability_1.5.1  lifecycle_1.0.4      StanHeaders_2.32.10
[13] processx_3.8.6      magrittr_2.0.4        compiler_4.5.2
[16] rlang_1.1.6         tools_4.5.2          utf8_1.2.6
[19] yaml_2.3.10         data.table_1.17.8    labeling_0.4.3
[22] bridgesampling_1.2-1 pkgbuild_1.4.8        plyr_1.8.9
[25] RColorBrewer_1.1-3  cmdstanr_0.9.0        abind_1.4-8
[28] withr_3.0.2         datawizard_1.3.0     grid_4.5.2
[31] stats4_4.5.2        xtable_1.8-4         inline_0.3.21
[34] ggribes_0.5.7       emmeans_2.0.1        scales_1.4.0
[37] tinytex_0.57        insight_1.4.4         cli_3.6.5
[40] mvtnorm_1.3-3       rmarkdown_2.30        generics_0.1.4
[43] RcppParallel_5.1.11-1 reshape2_1.4.5        tzdb_0.5.0
[46] rstan_2.32.7        HDInterval_0.2.4     parallel_4.5.2
[49] matrixStats_1.5.0   vctrs_0.6.5          Matrix_1.7-4
[52] jsonlite_2.0.0      hms_1.1.4            arrayhelpers_1.1-0
[55] ggdist_3.3.3        see_0.12.0           glue_1.8.0
[58] codetools_0.2-20    ps_1.9.1             distributional_0.5.0
[61] stringi_1.8.7       gtable_0.3.6         QuickJSR_1.8.1
[64] pillar_1.11.1       htmltools_0.5.8.1    Brodingtonag_1.2-9
[67] R6_2.6.1            evaluate_1.0.5        lattice_0.22-7
[70] backports_1.5.0     rstantools_2.5.0     coda_0.19-4.1
[73] gridExtra_2.3       nlme_3.1-168         checkmate_2.3.3
[76] xfun_0.55           pkgconfig_2.0.3
```